SOFTWARE REQUIREMENTS SPECIFICATION

for

ICU MULTIMODAL PATIENT MONITORING SYSTEM

with Real-Time Face and Eye Behaviour Analysis

Version 1.0

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1. Introduction

1.1. Purpose

Intensive Care Units (ICUs) depend on continuous observation of patients. Physiological monitors, periodic charting and manual assessment remain the primary sources of information, but visual cues such as facial expression, eye closure and gaze behaviour are rarely captured in a structured, reviewable form. The team has already built a Python computer-vision pipeline (the *Face + Eye Pipeline*) that extracts such cues from a camera stream. At present its results are shown in a local OpenCV window and written to CSV/JSONL logs; the small FastAPI service in the same repository keeps only the latest values in memory, has no database, no authentication and no user interface for clinical staff.

This document is the Software Requirements Specification (SRS) for the **ICU Multimodal Patient Monitoring System**, Version 1.0: a web platform that connects the existing pipeline to a full-stack application (React + Vite, Node.js + Express, MongoDB Atlas) so that authorised users can view live face-and-eye observations, review history, receive configured event alerts and manage ICU patient records from one interface. The SRS defines what the system shall do and the limits within which it shall operate. It is the baseline for design, implementation, testing and acceptance.

The system is a **research and monitoring-support prototype**. It does not diagnose, does not recommend treatment and does not replace physiological measurement or clinical judgement.

1.2. Document Conventions

- **Requirement identifiers** follow the pattern `FR-<MODULE>-nn` (functional), `CVI-nn` (changes required in the existing CV pipeline), `NFR-<CATEGORY>-nn` (non-functional), `DR-nn` (data), `IR-<TYPE>-nn` (interface), `C-nn` (constraint), `A-nn` (assumption), `BR-nn` (business rule) and `R-nn` (risk or known issue).
- **Keywords.** "Shall" marks a mandatory requirement, "should" a recommended one and "may" an optional one.
- **Priority** uses MoSCoW: **M** = Must have (needed for the core demonstration), **S** = Should have, **C** = Could have (only if time permits). Items that are deliberately excluded are listed in Section 1.4.2.
- **Numeric targets** (latencies, timeouts, rates, retention periods) are initial design targets. They shall be confirmed with the supervisor and validated by testing before being quoted as achieved results.
- **Evidence about the CV pipeline.** Statements about the existing pipeline come from the project hand-off document, which is a static review of the public repository (branch `main`, commit `5405f41`, dated 2026-07-08). The pipeline has not been run end-to-end against the new platform; behaviour must be re-verified during integration (Phase 5 of the proposal).

1.3. Intended Audience and Reading Suggestions

This SRS is written for developers, the project supervisor and evaluators, testers, and any clinical or ethics adviser who reviews the project. It covers external, internal, functional and non-functional aspects of the system.

- **Supervisor / evaluators:** read Chapters 1 and 2, then Appendix A (traceability) and Section 5.5 (known issues).
- **Frontend developers:** Sections 2.8, 3.1 (features 1–3 and 7–14) and Appendix **??**.
- **Backend developers:** Sections 3.1 (features 2–9), 3.17, 3.19 and 3.21.
- **CV / pipeline developers:** Sections 3.17, 3.20 and 5.5.
- **Testers:** every requirement table in Chapter 3 and Chapter 4; each requirement is written to be testable.

1.4. Project Scope

The ICU Monitoring Platform creates a shared space for **administrators, doctors, nurses and family members** (Section 2.3) around one core capability: turning the output of the Face + Eye Pipeline into a persistent, access-controlled, reviewable record.

A camera is placed at the bed. The Python pipeline detects and tracks one patient face, estimates facial Action Units (AUs), head pose, eye closure, PERCLOS and gaze, and computes heuristic indices (pain-related facial behaviour, fear, fatigue, agitation, tension, a respiratory-distress proxy and a global distress score) together with episode flags. It sends a feature payload (and, optionally, a live preview frame) over HTTPS to the application backend. The backend validates and stores the data in MongoDB Atlas and serves it to role-specific dashboards. Around this core, the platform provides patient and monitoring-session management, an alert centre, historical analytics and the administrative workspaces that appear in the supplied reference video (staffing, rooms, shifts, tasks and medication records).

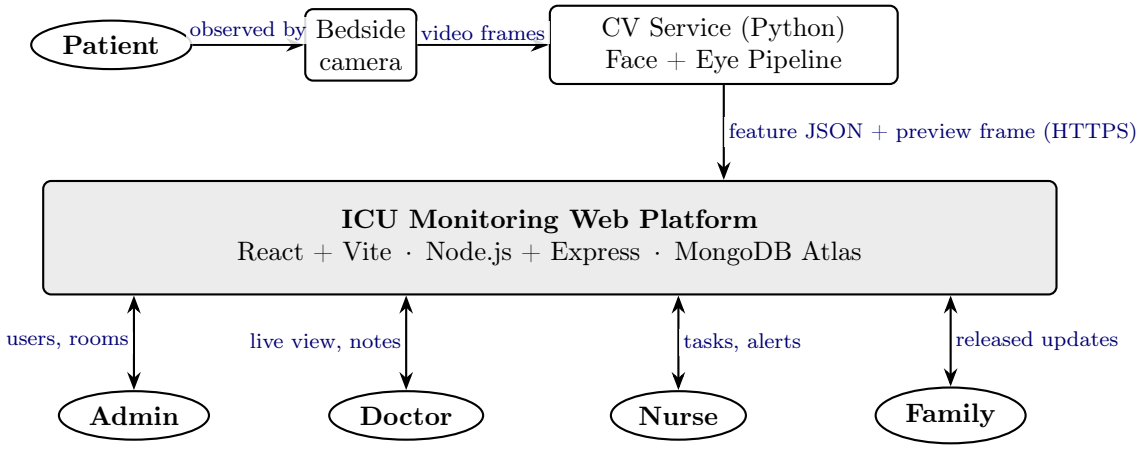


Figure 1.1.: System context: the existing CV pipeline feeds the new web platform, which serves four user roles.

Figure 1.1 gives the overview. The connections between the entities are interdependent: the camera feeds the pipeline, the pipeline feeds the platform, and every user role works only through the platform.

1.4.1. In Scope

- Web application with landing page, workspace selection, role-based authentication and four workspaces (Admin, Doctor, Nurse, Family).
- Patient records, monitoring sessions and room/bed allocation.
- Versioned REST API for ingestion of face-and-eye feature payloads and preview frames from the Python service.
- Live (near-real-time) face-and-eye monitoring view with explicit stale, offline and missing-data states.
- Monitoring events, an alert centre with acknowledgement, and historical charts for each session.
- Administrative record keeping for care tasks, medication schedules, shifts and staff activity.
- Deployment on Vercel (frontend), Render (backend) and MongoDB Atlas (database); the CV service runs on a separate machine with the camera.

1.4.2. Out of Scope

The reference video shows several capabilities that this version does *not* implement. They shall not appear in the interface with invented data (BR-03, FR-LIVE-16).

- Clinical diagnosis, treatment advice, automatic medication decisions or drug-interaction checking.
- Acquisition of physiological vitals (heart rate, SpO_2, blood pressure, core temperature) and the "AI risk prediction" widgets (e.g. sepsis probability), lab results and a seizure score.
- The posture (pose) and audio analysis modules (the posture module is separate in the repository).
- Equipment inventory, billing, and integration with hospital information systems (HL7, FHIR, EHR).
- Permanent video recording, native mobile applications and clinical validation of the CV indices.

1.5. Project Objectives

ID	Objective
O1	Integrate the existing Face + Eye Pipeline through a documented, versioned HTTP contract, without re-implementing its algorithms.
O2	Provide secure, server-enforced role-based access for administrators, doctors, nurses and family members.
O3	Persist patients, monitoring sessions, feature snapshots, events and alerts in MongoDB Atlas.
O4	Show live face-and-eye observations within the latency target of NFR-PERF-01, and represent missing, stale or unavailable data explicitly.
O5	Convert pipeline episode flags into events and alerts that staff can review and acknowledge.
O6	Provide historical charts and session summaries based on recorded observations only.
O7	Support the operational ICU modules shown in the reference video (rooms, shifts, tasks, medication records) as administrative record keeping.
O8	Deliver a responsive interface modelled on the supplied reference video.
O9	Deploy frontend, backend and database to the chosen cloud services and connect them securely to the CV service.
O10	Test the complete data flow and document the architecture, limitations and future work, without presenting heuristic scores as clinically validated.

1.6. Definitions, Acronyms and Abbreviations

Term	Meaning
AU	Facial Action Unit; here a continuous geometric score estimated from facial landmarks (15 AUs), not a validated FACS intensity.
Alert	A notification raised by the platform from an event or rule and shown in the alert centre until acknowledged.
Baseline	Per-patient reference values collected by the pipeline during an initial period (about 30 s) before personal deviations are used.
Behaviour state	Rule-based label from the pipeline: `alert`, `drowsy`, `eye_closed`, `fixation` or `scanning`.
CV Service	The Python Face + Eye Pipeline running on the machine with the camera.
EAR	Eye Aspect Ratio, used to estimate eye closure.
Episode / Event	A period during which a configured threshold persists (pain, agitation, fatigue, fear, eye closure). The pipeline calls it an *episode*; the platform stores it as a *monitoring event*.
Global distress	Weighted heuristic score from 0 to 100 combining the pain, fear, fatigue, agitation, tension and respiratory indices.
Heuristic index	A score computed by fixed rules from visual features; not a direct physiological or clinical measurement.
L2CS-Net	Pretrained neural network used by the pipeline for gaze yaw and pitch.
MediaPipe Face Mesh	Library used for face detection and 468-point landmark extraction.
PERCLOS	Percentage of eye closure over a rolling time window.
RBAC	Role-Based Access Control.
REST / JSON	Representational State Transfer API style; JavaScript Object Notation payloads.

Term	Meaning
JWT	JSON Web Token used for authentication.
SPA	Single-Page Application (the React frontend).
Session	A monitoring session: the period during which a CV Service streams data for one patient.
Snapshot	One stored feature payload (a time-stamped set of face-and-eye values).
Stale data	The latest snapshot is older than the configured freshness threshold.
TTL index	MongoDB index that automatically deletes documents after a set age.

1.7. References

[1] Project Proposal, *ICU Multimodal Patient Monitoring System with Real-Time Face and Eye Behaviour Analysis*.

[2] *Face + Eye Pipeline — Complete Technical Handoff* (`FACE_EYE_PIPELINE_HANDOFF.md`), prepared 2026-10-16.

[3] Source repository: https://github.com/krishgakhar/2D_FaceAndEye_Pipline (branch `main`, commit `5405f41`).

[4] Supplied ICU monitoring interface reference video (landing page, workspace selection, doctor, nurse and admin screens).

[5] ISO/IEC/IEEE 29148:2018, *Systems and software engineering — Life cycle processes — Requirements engineering*; IEEE Std 830-1998, *Recommended Practice for Software Requirements Specifications*.

[6] SRS template (Overleaf), *Software Requirements Specification — IICT Website*, used for the chapter structure of this document.

2. Overall Description

2.1. Product Perspective

The platform is a new, self-contained web system that *wraps* an existing component. It does not replace the Face + Eye Pipeline; it replaces the pipeline's temporary delivery path (local window, CSV/JSONL files and the in-memory FastAPI prototype exposing `/face`, `/posture` and `/fusion`) with a persistent, authenticated, multi-user service. Three layers cooperate (Figure 2.1):

1. **Frontend** — React + Vite single-page application on Vercel.
2. **Application backend** — Node.js + Express REST API on Render, with MongoDB Atlas as database. The backend never executes the Python pipeline inside its own process.
3. **CV Service** — the existing Python pipeline on a machine that has the camera. It pushes validated payloads to the backend.

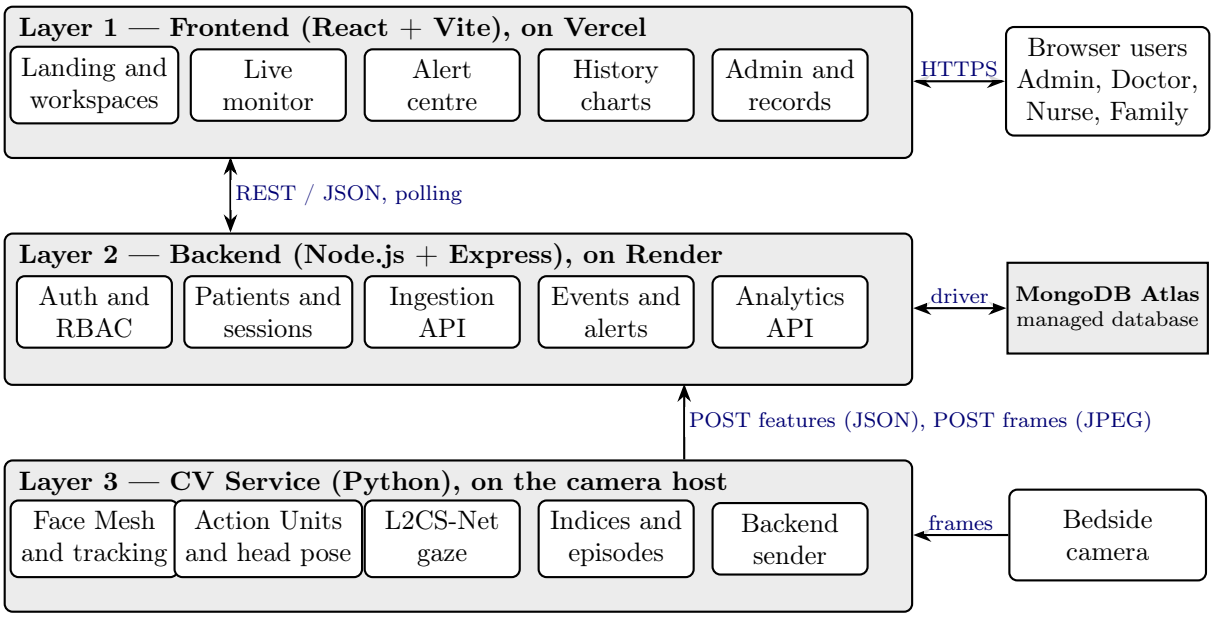


Figure 2.1.: Layered architecture and main data flow: camera → CV Service → backend → MongoDB Atlas → React dashboard.

2.2. Stakeholders

Table 2.1.: Stakeholders and their concerns

Stakeholder	Role	Interest and main concerns
Doctors (intensivists)	Primary users of the live monitor, alert centre and history.	Timely, understandable signals; avoiding alert fatigue; not over-interpreting heuristic scores.
ICU nurses	Bedside users; acknowledge alerts, record tasks and medication administration.	Fast, low-effort screens during care; clear stale or offline indicators.
ICU / hospital administrators	Manage users, rooms, staffing and system status.	Correct access control, audit trail, system availability.
Patients	Subjects observed by the camera.	Privacy, dignity and consent; no unnecessary storage of video; minimum personal data.
Family / guardians	Receive limited, released updates.	Understandable information without exposure of raw monitoring data.
CV / ML developers	Own the Face + Eye Pipeline and its payload.	A stable, versioned contract; sender changes that do not stall the camera loop.
Web and backend developers	Build and deploy the platform.	Clear API contract, testable requirements, reproducible deployment.
Supervisor and evaluators	Assess the project.	Traceability, honest statement of limitations, evidence from testing.
Ethics / clinical adviser	Approves any use with real patients or volunteers.	Consent, data minimisation, no unvalidated clinical claims.
Cloud providers (Vercel, Render, MongoDB Atlas)	External services on which the system depends.	Plan limits, idle behaviour, credentials and network rules.

2.3. User Classes and Characteristics

The platform has four human user classes and one system actor (Figure 2.2). Staff accounts are created by an administrator; there is no public self-registration.

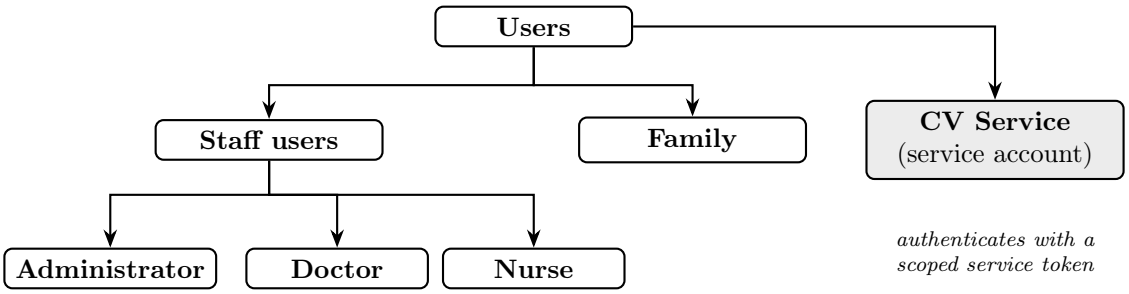


Figure 2.2.: Types of users.

Table 2.2.: User classes

Class	Description	Typical characteristics
Administrator	Manages user accounts, rooms and beds, staffing views, system status and the audit log. Does not use the live clinical view.	Operations staff; works at a desk; needs overview screens and strict control of who can see what.
Doctor	Reviews assigned patients, the live monitor, alerts and history; writes clinical notes; creates tasks and medication records; releases status notes to family.	High clinical expertise, limited time; mostly desktop or tablet; needs concise screens.
Nurse	Checks in to a shift, watches assigned patients, acknowledges alerts, completes care tasks, records medication as given.	Frequent short interactions at the bedside, often on a tablet; needs large touch targets.
Family member	Read-only access to a limited recovery summary of one linked patient.	Non-technical; may use a phone; must never see raw indices, video or other patients.
CV Service	Non-human client that posts feature payloads, preview frames and heartbeats.	Runs unattended; authenticates with a revocable service token; has no access to user-facing data.

Table 2.3 fixes the permissions that the backend shall enforce (FR-WS-03).

Table 2.3.: Role permission matrix. RW = read and write, R = read-only, (a) = assigned patients only, L = limited to released information of the linked patient, – = no access.

Capability	Admin	Doctor	Nurse	Family
Sign in and select workspace	RW	RW	RW	RW
Create / edit patient records	RW	RW	R(a)	–
View patient list and details	R	R(a)	R(a)	L
Start / end monitoring session	–	RW(a)	RW(a)	–
Live face-and-eye monitor	–	R(a)	R(a)	–
Alert centre (view, acknowledge)	R	RW(a)	RW(a)	–
Patient-level history and charts	–	R(a)	R(a)	–
Aggregate analytics (no identities)	R	–	–	–
Clinical notes	–	RW(a)	R(a)	–
Care tasks	R	RW(a)	RW(a)	–
Medication schedule (record keeping)	–	RW(a)	RW(a)	–
Shift check-in / check-out	–	RW (own)	RW (own)	–
Rooms and bed allocation	RW	R	R	–
Staffing and staff activity	R	–	–	–
User management and credentials	RW	–	–	–
Audit log and system status	R	–	–	–
Messaging	RW	RW	RW	L
Released status updates	–	RW(a)	–	R

2.4. Product Functions

The platform consists of the following functional modules (details in Chapter 3):

1. Landing page, authentication and workspace selection.
2. Patient and monitoring-session management.
3. CV payload ingestion (features, heartbeat and preview frames).
4. Live face-and-eye monitoring.
5. Monitoring events and the alert centre.
6. Historical analytics.
7. Doctor, nurse, administrator and family workspaces, including rooms, shifts, tasks, medication records and messaging.
8. Audit logging and system status.

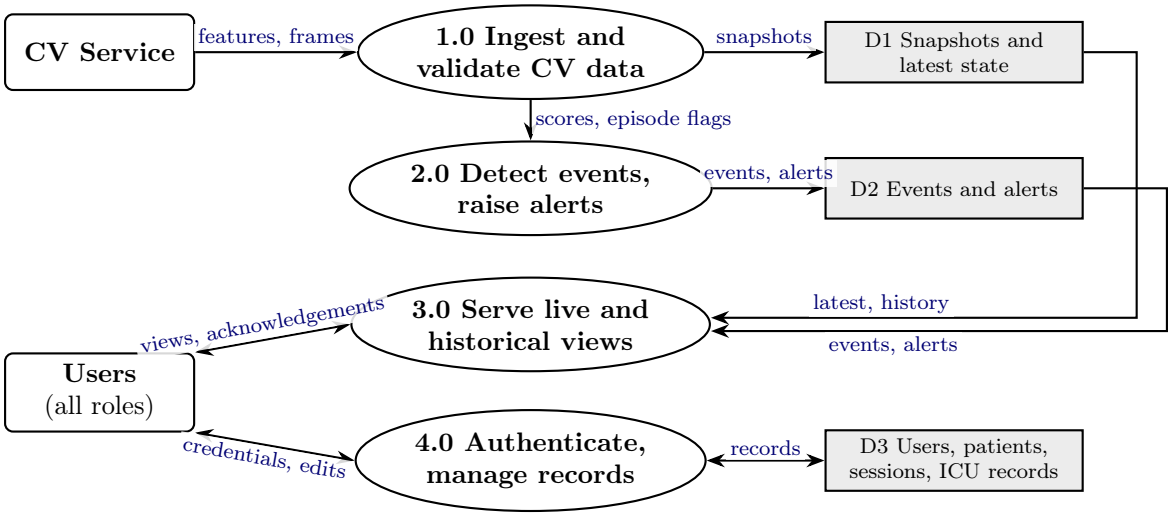


Figure 2.3.: Data flow diagram (Level 1).

2.5. Operating Environment

- **Client:** current versions of Chrome, Edge, Firefox and Safari on desktop, tablet and phone (viewport width from 360 px to 1920 px). No plug-ins are required.
- **Application backend:** Node.js (an active LTS release, fixed in `package.json`) with Express, hosted on Render.
- **Database:** MongoDB Atlas managed cluster.
- **CV Service:** Python 3.10 (as recommended in the repository README) on Windows or Linux, with a webcam or RGB camera, MediaPipe, OpenCV, PyTorch and the local weights file `models/L2CSNet_gaze360.pkl` (about 95.8 MB). A CUDA GPU is optional; a Jetson build uses `requirements/jetson.txt`.
- **Network:** the CV Service host needs outbound HTTPS access to the backend. No inbound connection to the CV host is required.

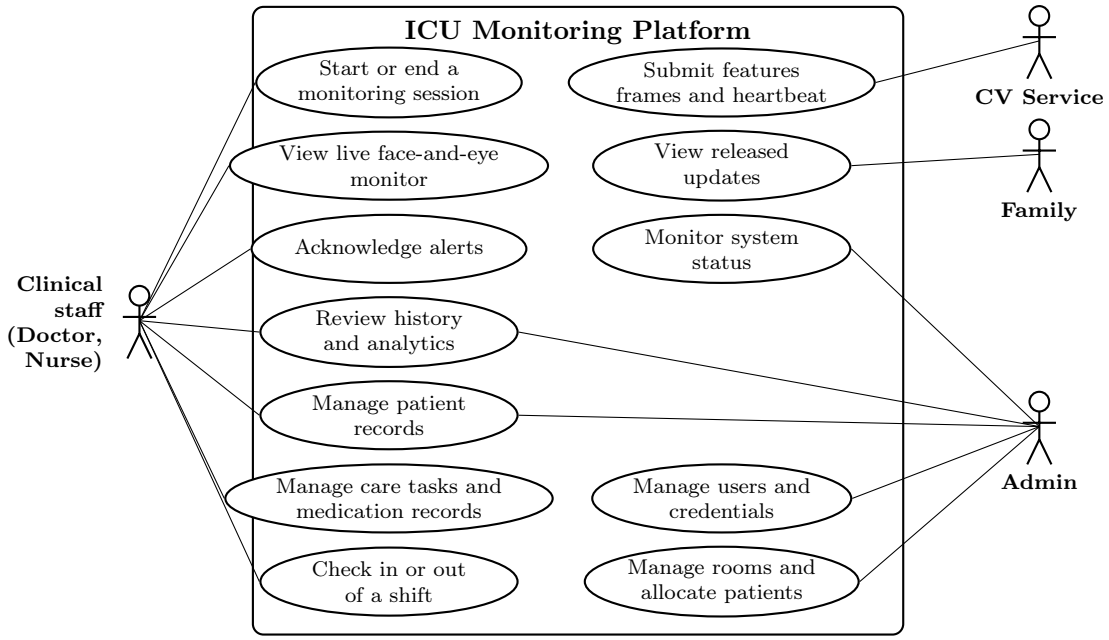


Figure 2.4.: Use-case diagram. Every human actor authenticates first (FR-AUTH); this is omitted for clarity. Doctor-only and nurse-only restrictions follow Table 2.3.

2.6. Design and Implementation Constraints

Table 2.4.: Constraints

ID	Statement
C-01	The frontend shall use React with Vite, the backend Node.js with Express, the database MongoDB Atlas; hosting shall use Vercel (frontend) and Render (backend). Source control uses Git and GitHub.
C-02	The Python pipeline shall remain a separate process. The Node.js backend shall not execute it in-process; they communicate only through the HTTP contract of Section 3.19.
C-03	A cloud-hosted website cannot reach a camera attached to a developer machine. The CV Service must therefore run where the camera is and push data to the cloud API.
C-04	The pipeline needs PyTorch, MediaPipe and a 95.8 MB weights file, and ideally a GPU. It shall not be assumed to fit the free tiers of Vercel or Render; the hosting decision for the CV Service is deferred (Appendix C).
C-05	The current sender posts feature payloads synchronously with a 1 s timeout from inside the camera loop. A slow backend therefore stalls inference; the ingestion endpoint must answer quickly (NFR-PERF-02) and the sender must be changed (CVI-03).
C-06	Preview frames are sent by a daemon thread that keeps only the latest frame with a 0.2 s timeout. Streaming is best-effort and lossy; it is a live preview, not an archive.
C-07	The pipeline scores one tracked face per camera. One CV instance therefore monitors one patient at a time, although MediaPipe may detect up to four faces.
C-08	All indices, behaviour states, severity bands and episodes are heuristics with software-defined thresholds. They are not clinically validated and shall be described as such in the interface and documents.

ID	Statement
C-09	In the current code path gaze yaw and pitch are in radians. The contract shall state units explicitly; the interface converts to degrees for display.
C-10	The payload now sent to the backend is a subset of the data computed by the pipeline (no AUs, head-pose angles, PERCLOS, mean/max gaze speed or baseline flag). Displaying them requires a deliberate payload extension (CVI-02).
C-11	The current pipeline emits a payload only while a face is tracked and sends no heartbeat. Without a heartbeat the platform cannot distinguish "no face in view" from "CV Service stopped" (CVI-06).
C-12	Hosting plans may spin down idle services and limit database storage. Data volume shall be controlled by rate limiting, down-sampled history and TTL expiry (FR-ING-05, DR-05).
C-13	Patient data shall be limited to the minimum needed for the project, and camera frames shall not be stored permanently by default (NFR-PRV-01, NFR-PRV-02).

2.7. Assumptions and Dependencies

Table 2.5.: Assumptions and dependencies

ID	Statement
A-01	The pipeline behaviour described here matches the audited repository state (commit `5405f41`). Any divergence found during integration is recorded and this SRS is updated.
A-02	A monitoring session is bound to one patient. The `patient_id` sent by the CV Service (default `P001`) is mapped to a platform patient record by the person who starts the session.
A-03	The pipeline baseline period (about 30 s from start or after reset) is assumed to show a neutral face. This is an operating assumption, not something the system can verify.
A-04	The CV host and the backend have reasonably synchronised clocks. The backend stores both the capture time and its own receive time and uses the receive time for ordering.
A-05	The pipeline sender will be modified as specified in Section 3.17. Until then it will not work with the new API: it is disabled by default (`ENABLE_BACKEND = False`) and its routes differ from the contract.
A-06	Vercel, Render and MongoDB Atlas remain available under the plan selected for the project.
A-07	Testing uses volunteers, recorded or synthetic data. Use with real patients requires prior ethics approval and consent.

2.8. User Interface Design

The interface is modelled on the supplied reference video: a clean light theme with a blue accent, a public landing page, a "Select Your Workspace" page, one login portal per role, and a left-hand navigation rail inside each workspace. Appendix **??** reproduces the reference screens. Table 2.6 lists the screens to be built.

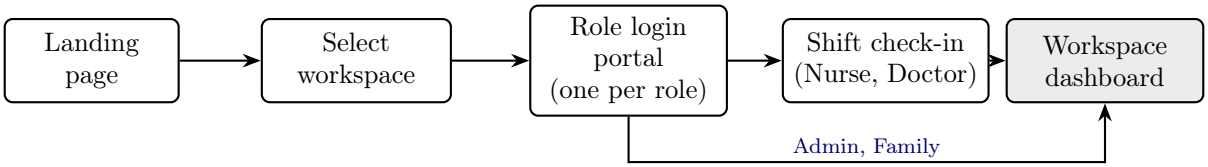


Figure 2.5.: Entry flow shared by all workspaces.

Admin	Doctor	Nurse	Family
Staffing and rooms	Roster	Shift check-in	Recovery summary
Patients	Currently monitoring	Overview	Released updates
Department view	Live patient monitor	Care tasks	Messages
Analytics (aggregate)	Alert centre	Medication schedule	
User management	Clinical notes	Live monitor (assigned)	
System status	Tasks and medication	Alerts	
Audit log	Shift handoff	Check-out	
Messages			

Figure 2.6.: Workspace activities by role.

Table 2.6.: Screen inventory derived from the reference video

Screen	Main content	Features
Landing page	Hero text, platform / how-it-works / safety sections, sign-in and request-access buttons, prototype disclaimer.	F1
Select Your Workspace	Four tiles: Admin (System Ops), Doctor (Clinical Rounds), Nurse (Bedside Care), Family (My Recovery).	F3
Role login portals	Admin Portal, Doctor Portal, Nurse Station: ID or e-mail, password with show toggle, sign-in button, forgot-credentials link.	F2
Start of shift dialog	Nurse name, shift selector (e.g. Morning 07:00–15:00), check-in button.	F11
Doctor dashboard	Header with shift status, emergency call and request handoff; active roster; "currently monitoring" panel with distress score; alert centre.	F10, F8

Screen	Main content	Features
Live Patient Monitor	Patient roster, distress score and alert level, Face & Eye panel with preview and values, freshness indicator. Posture and audio panels are not built.	F7
Nurse overview, care tasks, medication schedule	Assigned beds, pending medication, tasks due, last check; task list; scheduled medication with "Given" status.	F11
Admin dashboard	Tabs for staffing, patients, department, analytics, user management, credentials and messages; floor and room management with allocate, deallocate and set-critical; alerts; active doctors; nurse activity dialog.	F12, F14
Messages	Contact list and thread view, including system messages for shift check-in and check-out.	F14
Family view	Read-only recovery summary of the linked patient.	F13

3. System Features

The ICU Monitoring Platform is a monitoring-support and record-keeping web system. Its main value is to turn the output of the Face + Eye Pipeline into stored, access-controlled and reviewable information; every other feature supports that flow.

3.1. Description and Priority

The features are listed from highest to lowest priority. All **Must** features are needed for the core demonstration; **Should** and **Could** features follow the reference video and are built if time allows.

Table 3.1.: Feature list, module codes and priorities

No.	Feature	Module	Reference	Priority
F6	CV payload ingestion	ING	Proposal §9; hand-off §9	Must
F7	Live face-and-eye monitoring	LIVE	Live Patient Monitor screen	Must
F2	Authentication	AUTH	Role login portals	Must
F3	Workspace selection and access control	WS	Select Your Workspace	Must
F4	Patient management	PAT	Roster, patient cards	Must
F5	Monitoring-session management	SESS	Proposal §10	Must
F8	Monitoring events and alert centre	EVT	Alert Center	Must
F9	Historical analytics	HIST	Proposal §8.7	Must
F10	Doctor workspace	DOC	Doctor dashboard	Should
F11	Nurse workspace	NUR	Nurse overview, tasks, medication	Should
F12	Administrator workspace	ADM	Admin dashboard	Should
F1	Public landing page	LAND	Landing page	Should
F15	Audit logging	AUD	Proposal §12	Should
F13	Family workspace	FAM	My Recovery	Could
F14	Messaging, emergency call, handoff	MSG	Messages, header buttons	Could

3.2. Feature 1: Public Landing Page

The landing page introduces the platform to visitors and routes them to sign-in. It describes the system honestly as a research prototype.

Table 3.2.: Landing page requirements

ID	Requirement	Pri.
FR-LAND-01	The system shall provide a public landing page that explains the platform's purpose and main capabilities and links to sign-in and workspace selection.	S
FR-LAND-02	The landing page shall state that the system is a research and monitoring-support prototype and not a diagnostic or treatment device.	M
FR-LAND-03	Any illustrative patient card on the landing page shall be labelled as sample data; it shall not appear as a live feed.	M
FR-LAND-04	The landing page shall not claim regulatory or security certifications (for example HIPAA or SOC 2) that the project has not obtained.	M

3.3. Feature 2: Authentication

Only registered users can enter the system. Accounts are created by an administrator, and the CV Service has its own credential.

Table 3.3.: Authentication requirements

ID	Requirement	Pri.
FR-AUTH-01	The system shall let a registered user sign in with an e-mail or staff ID and a password and sign out.	M
FR-AUTH-02	Passwords shall be stored only as salted hashes produced by an adaptive algorithm (bcrypt or Argon2id) and shall never be logged or returned.	M
FR-AUTH-03	On successful sign-in the backend shall issue a signed, expiring token (JWT in an HttpOnly, Secure cookie, or an equivalent secure session) that identifies the user and role.	M
FR-AUTH-04	The endpoint `GET /api/v1/auth/me` shall return the current user's identity and role, or 401 if the token is missing, invalid or expired.	M
FR-AUTH-05	User accounts shall be created only by an administrator; public self-registration shall be disabled.	M
FR-AUTH-06	Clinical workspaces shall end the session after a configurable idle period (default 30 minutes) and require sign-in again.	S
FR-AUTH-07	The backend shall limit repeated failed sign-ins (default: temporary lock after 5 failures within 10 minutes) and return a generic error that does not reveal whether the account exists.	S
FR-AUTH-08	An administrator shall be able to reset a user's credentials (the "forgot credentials" path); no e-mail service is required.	C
FR-AUTH-09	The CV Service shall authenticate with a separate, revocable service token that is accepted only by the ingestion endpoints (see NFR-SEC-07).	M

3.4. Feature 3: Workspace Selection and Access Control

After the landing page the user chooses a workspace (Admin, Doctor, Nurse or Family). The backend decides what each role may do (Table 2.3); the interface only reflects that decision.

Table 3.4.: Workspace selection and access-control requirements

ID	Requirement	Pri.
FR-WS-01	The system shall provide a "Select Your Workspace" page with one tile each for Admin, Doctor, Nurse and Family, each leading to its login portal.	M
FR-WS-02	After sign-in the backend shall check that the user's role matches the selected workspace and reject a mismatch with a generic error.	M
FR-WS-03	Every protected endpoint shall enforce the permissions of Table 2.3 on the server, including object-level checks such as "assigned patients only". Hiding a control in the frontend is not sufficient.	M
FR-WS-04	The frontend shall redirect unauthenticated users to the workspace-selection page and shall not render a workspace for a role the user does not hold.	M
FR-WS-05	Unauthorised requests shall receive 401 (not authenticated) or 403 (not permitted) with the standard error body of Section 3.19.	M

3.5. Feature 4: Patient Management

Patient records hold the minimum data needed to link monitoring data to a person, assign staff and allocate a bed.

Table 3.5.: Patient-management requirements

ID	Requirement	Pri.
FR-PAT-01	Authorised users shall be able to create a patient record with only the minimum fields: a unique patient reference, a display name or alias, age, admission date, an optional free-text admission reason, the assigned doctor, assigned nurse(s) and bed.	M
FR-PAT-02	Authorised users shall be able to update a patient record; every update shall record who changed it and when.	M
FR-PAT-03	The system shall list patients with search and filters (name, reference, bed, status) and shall show only the patients the user is allowed to see (Table 2.3).	M
FR-PAT-04	The patient detail page shall show the record, current session status, recent events, tasks and medication entries, with links to the live monitor and history.	M
FR-PAT-05	The patient reference shall equal the `patient_id` sent by the CV Service (default `P001`) so that payloads can be matched to the record.	M
FR-PAT-06	The system shall not collect government identifiers, home addresses, photographs or insurance data.	M
FR-PAT-07	`DELETE /api/v1/patients/:patientId` shall be restricted to administrators and shall archive the record (soft delete) instead of removing it; its monitoring history shall remain available to authorised users.	S

ID	Requirement	Pri.
FR-PAT-08	An administrator or doctor shall be able to assign and change the doctor and nurse(s) responsible for a patient.	M

3.6. Feature 5: Monitoring-Session Management

A monitoring session ties one patient to one CV Service source for a period of time. Its status follows Figure 3.1.

Table 3.6.: Monitoring-session requirements

ID	Requirement	Pri.
FR-SESS-01	A doctor or nurse shall be able to start a monitoring session for a patient; the system shall record the patient, the starting user, the CV source identifier and the start time.	M
FR-SESS-02	The system shall allow at most one active session per patient and per CV source.	M
FR-SESS-03	A doctor or nurse shall be able to end a session; the system shall record the end time.	M
FR-SESS-04	The session status shall be derived by the backend: *active* while valid payloads or heartbeats arrive, *offline* when none has arrived for the offline threshold (default 30 s), and *ended* after it is closed.	M
FR-SESS-05	The system shall list the sessions of a patient with start, end, duration and counts of events by type.	M
FR-SESS-06	When a payload arrives for a patient with no active session, the backend shall reject it with 409 and the error code NO_ACTIVE_SESSION; it shall not create sessions silently.	M
FR-SESS-07	The session shall store the payload schema version, the pipeline version (when supplied) and the last baseline-ready flag reported by the CV Service.	S

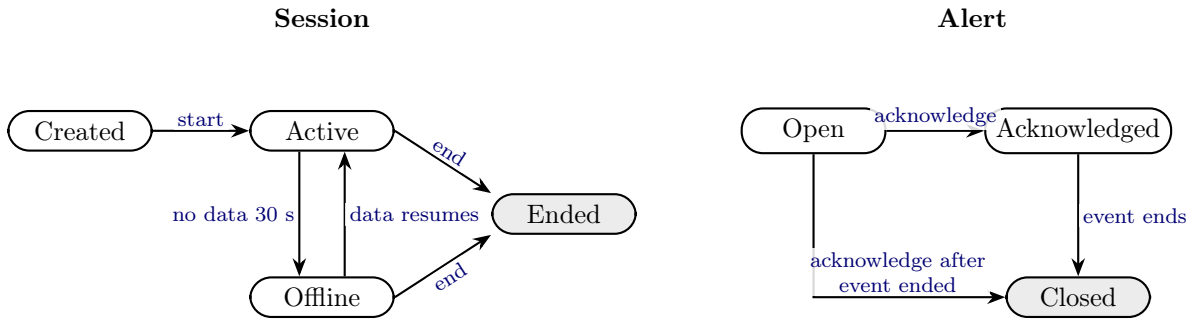


Figure 3.1.: State diagrams for a monitoring session and an alert.

3.7. Feature 6: CV Payload Ingestion

The ingestion API is the contract between the Python pipeline and the platform. The payload fields are defined in Section 3.20; the changes required in the pipeline are in Section 3.17. Figure 3.2 shows the nominal flow.

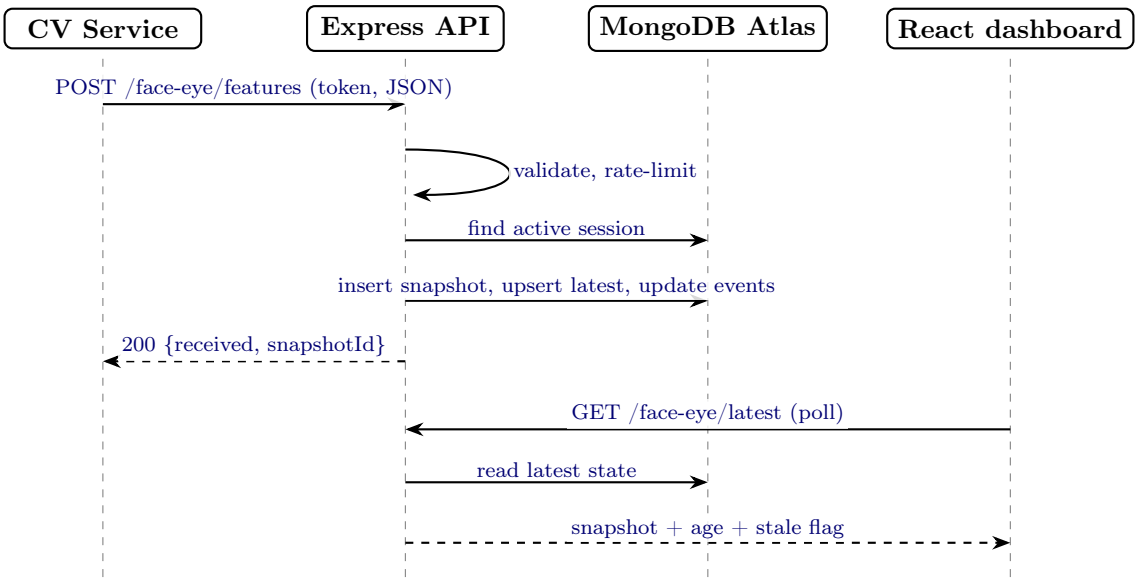


Figure 3.2.: Sequence of a feature payload from the CV Service to the dashboard.

Table 3.7.: Ingestion requirements

ID	Requirement	Pri.
FR-ING-01	`POST /api/v1/face-eye/features` shall accept a JSON feature payload from an authenticated CV Service.	M
FR-ING-02	The backend shall validate required fields, types and ranges (Table 3.20) and reject invalid payloads with 400 or 422 and a list of field errors. An invalid payload shall never be stored as valid.	M
FR-ING-03	The backend shall attach the matching active session, store the capture time (`timestamp_unix`) and its own receive time, and use the receive time for ordering.	M
FR-ING-04	The endpoint shall acknowledge a valid payload quickly with `{received, snapshotId, sessionId}` (target in NFR-PERF-02), because the current sender blocks for up to 1 s per request.	M
FR-ING-05	The latest state shall be updated for every accepted payload, but the history shall be stored at a configurable rate (default one snapshot per second per session) and additionally at every event start and end.	M
FR-ING-06	The latest monitoring state shall be kept separate from historical snapshots (see DR-03).	M
FR-ING-07	`GET /api/v1/face-eye/latest` shall return the latest snapshot for a patient together with its age in milliseconds and an `isStale` flag computed from the freshness threshold (default 5 s).	M
FR-ING-08	`GET /api/v1/face-eye/history` shall return snapshots for a patient or session, filtered by time range and fields, in ascending time order, with a maximum point count (default 2,000) and optional bucket averaging.	M

ID	Requirement	Pri.
FR-ING-09	Payloads shall carry `schema_version`. A payload without it shall be accepted as the legacy version `1.0`; unknown optional fields shall be ignored; an unsupported major version shall be rejected.	M
FR-ING-10	The backend should tolerate duplicate deliveries (same session and sequence number or timestamp) without creating duplicate snapshots.	S
FR-ING-11	`POST /api/v1/face-eye/frames` shall accept a multipart upload with the JPEG field `frame`; it shall check content type, magic bytes and size (default $\leq$ 1 MB) and keep only the latest frame per session in short-lived storage.	S
FR-ING-12	`GET /api/v1/face-eye/frames/latest` shall return the latest frame to authorised users only, with `Cache-Control: no-store`, or 404 when none exists or it is older than the freshness threshold.	S
FR-ING-13	Frames shall not be written to permanent storage. Any retention feature shall be disabled by default and shall require a documented purpose, access rule and retention period.	M
FR-ING-14	The backend shall store gaze angles in the unit declared by the payload (default radians for version 1.0); conversion to degrees is a display concern.	M
FR-ING-15	The ingestion endpoints shall limit request size (default 16 KB for features) and rate (default 5 requests per second per session) and answer 413 or 429 when exceeded.	S
FR-ING-16	`POST /api/v1/face-eye/heartbeat` shall accept a small status message (camera state, face detected, frame rate, baseline ready, pipeline version) and update the session's CV status.	S
FR-ING-17	`GET /api/v1/system/health` shall report API status, database connectivity and, for administrators, the age of the last payload per active session.	M

3.8. Feature 7: Live Face and Eye Monitoring

This is the core screen. It shows what the pipeline produced, how fresh it is and how reliable it is; it never fills gaps with invented values.

Table 3.8.: Live-monitor requirements

ID	Requirement	Pri.
FR-LIVE-01	The live monitor shall be patient-specific and show the patient, bed, session status and CV Service status.	M
FR-LIVE-02	The monitor shall show the latest preview frame with the pipeline's annotations when available, and a "No video" placeholder otherwise.	S
FR-LIVE-03	The monitor shall show the global distress score (0–100) with its severity label: Normal below 20, Mild below 40, Moderate below 60, Severe below 80, Critical otherwise. The thresholds shall be configurable and default to those of the pipeline.	M
FR-LIVE-04	The monitor shall show the pain, fear, fatigue, agitation, tension and respiratory indices (0–100 each). The respiratory index shall be labelled as a visual proxy and not a measured respiration.	M

ID	Requirement	Pri.
FR-LIVE-05	The monitor shall show the behaviour state (`alert`, `drowsy`, `eye_closed`, `fixation`, `scanning`).	M
FR-LIVE-06	The monitor shall show gaze yaw and pitch in degrees, gaze variance, fixation duration in seconds, recent saccade count and eye-contact ratio.	M
FR-LIVE-07	When the behaviour state is `eye_closed`, gaze variance and gaze speed shall be shown as "n/a (eyes closed)", because the gaze tracker returns zeroed variance and speed fields in that branch.	M
FR-LIVE-08	The monitor shall show face confidence as a feature-quality measure and not as the probability that a clinical interpretation is correct.	M
FR-LIVE-09	The monitor shall show active episode indicators for pain, agitation, fatigue, fear and eye closure and the time of the latest update.	M
FR-LIVE-10	When the payload provides them, the monitor shall also show the 15 AU values, head-pose yaw, pitch and roll, motion energy, asymmetry, PERCLOS and mean and maximum gaze speed. Fields that are absent shall read "Not provided".	S
FR-LIVE-11	The monitor shall refresh automatically by polling at a configurable interval (default 2 s). Server-Sent Events or WebSockets may replace polling later.	M
FR-LIVE-12	If the latest snapshot is older than the freshness threshold the monitor shall mark all values as *stale*, grey them out and show the last update time. If nothing has arrived for the offline threshold it shall show "CV Service offline". Missing values shall never be shown as zero.	M
FR-LIVE-13	When heartbeats are available, the monitor shall distinguish "no face detected" from "CV Service offline"; otherwise it shall show "No recent data".	S
FR-LIVE-14	The monitor shall show whether the baseline is ready; while it is not, values shall carry a "calibrating" note.	S
FR-LIVE-15	Users shall be able to switch between their rostered patients from the monitor.	M
FR-LIVE-16	The interface shall not display posture, audio, vitals (heart rate, SpO_2, blood pressure), seizure score, AI risk-prediction or lab panels. A placeholder, where used, shall read "Not available in this version".	M
FR-LIVE-17	Every monitoring screen shall show the notice "Research prototype — visual estimates are not a diagnosis".	M

3.9. Feature 8: Monitoring Events and Alert Centre

The pipeline's episode detector reports five episode types as flags. The platform turns each flag transition into a stored *event*, and selected events into *alerts* that staff acknowledge.

Table 3.9.: Event and alert requirements

ID	Requirement	Pri.
FR-EVT-01	When an episode flag (pain, agitation, fatigue, fear, eye closure) changes from false to true, the backend shall create an event with its start time; when it changes to false the backend shall close the event and record the end time and duration.	M
FR-EVT-02	Each event shall store type, patient, session, start, end, duration, the peak value of the related index and its status (active or completed).	M
FR-EVT-03	`GET /api/v1/face-eye/events` shall list events filtered by patient, session, type, status and time range.	M
FR-EVT-04	The backend shall raise an alert when an event of a configured type starts and when the global distress stays at or above the Severe threshold for a configured time (default 10 s). Rules are set in server configuration; editing rules in the interface is future scope.	M
FR-EVT-05	The alert centre shall list alerts newest first with severity, patient, bed, description, time and status, and shall keep unacknowledged alerts visible until they are acknowledged.	M
FR-EVT-06	A doctor or nurse responsible for the patient shall be able to acknowledge an alert; the system shall record the user and time.	M
FR-EVT-07	The backend should suppress repeated alerts of the same type for the same patient within a cool-down window (default 60 s).	S
FR-EVT-08	Alert text shall describe the configured visual rule that fired (for example “Pain-related facial behaviour episode started”) and shall not state a diagnosis.	M
FR-EVT-09	The interface may show a toast and an optional sound for new Severe or Critical alerts.	C
FR-EVT-10	The “Emergency call” button shall record a call request and notify on-duty staff in the platform. It shall not contact external emergency services.	C

3.10. Feature 9: Historical Analytics

Historical views let staff review how recorded observations changed over a session. They show recorded values and events only.

Table 3.10.: Historical-analytics requirements

ID	Requirement	Pri.
FR-HIST-01	Authorised users shall be able to select a patient, session and time range and see a time-series chart of the global distress score and of any single index.	M
FR-HIST-02	The history view shall list the events of the selected range with type, start, end and duration, aligned with the chart.	M
FR-HIST-03	Observations shall be ordered by timestamp. Gaps in data shall appear as gaps in the chart and shall not be interpolated or filled with zero.	M

ID	Requirement	Pri.
FR-HIST-04	The history view shall offer eye metrics (gaze variance, fixation duration, eye-contact ratio and, when provided, PERCLOS) and the share of time spent in each behaviour state.	S
FR-HIST-05	For each session the system shall show a summary: mean and maximum global distress, number of events by type, and data coverage (share of the session with fresh data).	S
FR-HIST-06	Long ranges shall be returned as bucket averages so that one response never exceeds the point limit of FR-ING-08.	S
FR-HIST-07	Authorised roles shall be able to export a session's snapshots and events as CSV; exports shall not contain video frames.	S
FR-HIST-08	Administrators shall see aggregate analytics (patient counts, sessions, events per day, occupancy) without patient identities.	S
FR-HIST-09	Charts shall be based on recorded observations only; the interface shall not present trends as diagnoses or predictions.	M

3.11. Feature 10: Doctor Workspace

The doctor workspace combines the roster, the currently monitored patient and the alert centre, as in the doctor dashboard of the reference video.

Table 3.11.: Doctor-workspace requirements

ID	Requirement	Pri.
FR-DOC-01	The doctor dashboard shall show the doctor's active roster (name, ward or bed, selection state) and a "currently monitoring" panel with the selected patient's latest distress score and status.	M
FR-DOC-02	The dashboard shall include the alert centre for the doctor's patients.	M
FR-DOC-03	The dashboard shall link to the live monitor and the history of each rostered patient.	M
FR-DOC-04	A doctor shall be able to add and edit timestamped clinical notes for an assigned patient; each note shall record its author.	S
FR-DOC-05	A doctor shall be able to create task and medication-schedule entries for assigned patients (record keeping only, see BR-04).	S
FR-DOC-06	A doctor shall be able to publish a short status note that the linked family account can read.	C
FR-DOC-07	A doctor shall be able to request a shift handoff, which creates a summary of open alerts, pending tasks and recent notes for the incoming doctor.	C

3.12. Feature 11: Nurse Workspace

The nurse workspace supports shift check-in, an overview of assigned beds and tasks, and the medication schedule.

Table 3.12.: Nurse-workspace requirements

ID	Requirement	Pri.
FR-NUR-01	A nurse shall check in to a shift (selected from the shifts defined by the administrator) before using the nurse station, and shall be able to check out; check-in and check-out times shall be stored.	S
FR-NUR-02	The nurse overview shall show assigned beds, pending medication, tasks due, time since the last recorded check, the nurse's patients and open alerts.	S
FR-NUR-03	A nurse shall be able to open the live monitor of each assigned patient and acknowledge alerts.	M
FR-NUR-04	A nurse shall be able to create, edit, complete and delete care tasks for assigned patients, each with a due time and status (for example "Wound care dressing change").	S
FR-NUR-05	A nurse shall be able to add scheduled medication entries (name, dose, route, patient, due time) and mark an entry as *Given*; the system shall record the user and time.	S
FR-NUR-06	Medication entries are records only: the system shall not recommend doses, check interactions or send orders (BR-04).	M
FR-NUR-07	Nurse login, check-in, check-out and completed tasks shall be visible to administrators as staff activity.	S

3.13. Feature 12: Administrator Workspace

The administrator workspace covers users, rooms, staffing and system status. It does not show live clinical video.

Table 3.13.: Administrator-workspace requirements

ID	Requirement	Pri.
FR-ADM-01	An administrator shall be able to create, edit and disable user accounts, assign roles, link family accounts to a patient and reset credentials.	M
FR-ADM-02	An administrator shall be able to define floors, rooms and beds and see each bed as available, occupied or critical.	S
FR-ADM-03	An administrator shall be able to allocate a patient to a free bed under a selected doctor, and to deallocate; the "critical" flag shall be a manual marker set by authorised staff and shall not be computed from the CV scores.	S
FR-ADM-04	The staffing view shall list nurses and doctors on duty with their shifts, and show a nurse's recent activity (login, check-in, tasks, check-out).	S
FR-ADM-05	The department view may group departments, doctors, nurses and their patients.	C

ID	Requirement	Pri.
FR-ADM-06	The system-status view shall show API health, database status and, for each active session, the CV Service status and the age of its last payload.	M
FR-ADM-07	The admin dashboard shall show the alert centre in a read-only form that links to the responsible clinicians.	S
FR-ADM-08	Equipment inventory shown in the reference video is out of scope for Version 1.0.	–

3.14. Feature 13: Family Workspace

The family workspace is a deliberately small, read-only view of information that a doctor has chosen to release.

Table 3.14.: Family-workspace requirements

ID	Requirement	Pri.
FR-FAM-01	A family account shall be linked by an administrator to one patient and shall see only that patient.	C
FR-FAM-02	The "My Recovery" view shall be read-only and show the patient's name, ward and the status notes released by a doctor.	C
FR-FAM-03	The family view shall not show raw indices, video frames, alerts, events or other patients' data.	C
FR-FAM-04	A family account shall be activated only after an administrator records that the patient or guardian has consented to family access.	C

3.15. Feature 14: Messaging, Emergency Call and Handoff

These features follow the messages tab and header buttons of the reference video. They are low priority.

Table 3.15.: Messaging requirements

ID	Requirement	Pri.
FR-MSG-01	Staff shall be able to send text messages to other staff accounts and read their threads; a family user shall be able to message the care team only.	C
FR-MSG-02	The system may post system messages for shift check-in and check-out, as shown in the reference video.	C
FR-MSG-03	Messages shall not be used to transmit CV frames or raw payloads.	C

3.16. Feature 15: Audit Logging

Audit logging records who did what, so that access to patient information can be reviewed.

Table 3.16.: Audit requirements

ID	Requirement	Pri.
FR-AUD-01	The system shall write an append-only audit entry (user, role, action, object, time, outcome) for sign-in and sign-out, role changes, patient-record access and edits, session start and end, alert acknowledgement and data export.	S
FR-AUD-02	Administrators shall be able to view and filter the audit log; no role shall be able to edit or delete entries through the interface.	S
FR-AUD-03	Audit entries shall not contain passwords, tokens, video frames or free-text clinical content.	S

3.17. CV Service Integration Requirements

The audit of the repository shows that the current pipeline cannot talk to the new API without changes. This section lists those changes. They are requirements on the *existing Python component* and are tracked as part of this project.

Table 3.17.: Required changes in the Face + Eye Pipeline

ID	Requirement	Pri.
CVI-01	Replace the hard-coded constants `ENABLE_BACKEND`, `BACKEND_URL` and `STREAM_URL` in `backend_sender.py` by configuration (environment variables) that point to `/api/v1/face-eye/features` and `/api/v1/face-eye/frames`. The checked-in values (`172.20.10.2:8000`, `/api/face-eye/features`, `/api/stream/face`) match neither the in-repo FastAPI service nor this contract.	M
CVI-02	Extend the payload as defined in Table 3.20: add `schema_version` and `units`; add the optional AUs, head pose, motion, EAR, PERCLOS, gaze-speed, baseline and frame-rate fields that the pipeline already computes.	M
CVI-03	Send payloads from a worker thread through a bounded queue, so that a slow or unreachable backend can no longer stall the camera loop. The drop policy (oldest first) and drop counters shall be documented.	M
CVI-04	Check the HTTP status of every request (for example with `raise_for_status`), log failures with counters, and describe delivery as best-effort. The current code has no retry, no durable queue and no delivery confirmation.	M
CVI-05	Send the service token in the `Authorization` header of every request.	M
CVI-06	Send a heartbeat every 5 s (camera state, face detected, frame rate, baseline ready, pipeline version) even when no face is tracked.	S
CVI-07	Limit feature posts to a configurable rate (default 2 per second, maximum 5) independent of the camera frame rate.	S
CVI-08	Limit preview frames to a configurable size and rate (default width $\leq$ 640 px, JPEG quality about 70, $\leq$ 2 frames per second) and add the `patient_id` form field to the multipart request.	S

ID	Requirement	Pri.
CVI-09	Compute gaze durations from timestamps or the measured frame rate instead of the fixed 30 FPS assumption, and reconcile the saccade-window comment (150 frames) with its stated meaning.	S
CVI-10	Use the configurable camera index of `CameraConfig` instead of the fixed device 0.	S
CVI-11	Align the JSONL log with the CSV and payload by adding gaze fields and the behaviour state.	C

3.18. External Interface Requirements

Table 3.18.: Interface requirements

ID	Statement
IR-UI-01	The frontend shall be a responsive React SPA that works from 360 px to 1920 px wide and follows the layout and navigation of the reference video (Section 2.8).
IR-UI-02	Colour shall not be the only carrier of meaning: status chips and severity shall also show text.
IR-HW-01	The CV Service shall read frames from a USB or built-in RGB camera (OpenCV `VideoCapture`) on the host machine.
IR-HW-02	The CV host shall be able to run MediaPipe Face Mesh and L2CS-Net inference; a CUDA GPU is optional.
IR-SW-01	The CV Service shall call the backend over HTTPS using JSON, plus multipart/form-data for preview frames.
IR-SW-02	The frontend shall call the backend over HTTPS using JSON. The backend shall allow cross-origin requests only from the configured frontend origin.
IR-SW-03	The backend shall access MongoDB Atlas through the official driver or Mongoose with a least-privilege database user.
IR-COM-01	All traffic shall use TLS. Timestamps in API responses shall be ISO 8601 in UTC; the CV payload carries `timestamp_unix` in seconds.

3.19. REST API Summary

All endpoints are under `/api/v1`. The list below extends the proposal's API with the endpoints needed by the features above; none of it exists yet and it is implemented during development.

- **Format:** JSON, UTF-8. Lists use `limit` and `cursor` parameters.
- **Errors:** `{ "error": { "code", "message", "details", "requestId" } }` with status 400, 401, 403, 404, 409, 413, 422, 429 or 503 as appropriate.
- **Authentication:** user token (cookie or bearer) for people; service token in `Authorization: Bearer` for the CV Service.
- **Roles:** P = public, A = Admin, D = Doctor, N = Nurse, F = Family, S = CV Service.

Table 3.19.: REST API summary

Method	Path	Purpose	Roles
POST	/auth/register	Create a user (admin only)	A
POST	/auth/login	Sign in	P
POST	/auth/logout	Sign out	all
GET	/auth/me	Current user and role	all
GET, PATCH	/users, /users/:userId	List users; edit, disable, reset	A
GET, POST	/patients	List, create patients	A, D, N(R)
GET, PATCH, DELETE	/patients/:patientId	Read, update, archive	A, D, N(R)
GET, POST	/patients/:patientId/sessions	List, start sessions	D, N
GET, PATCH	/sessions/:sessionId	Read, end a session	D, N
POST	/face-eye/features	Submit feature payload	S
POST	/face-eye/heartbeat	Submit CV status	S
POST	/face-eye/frames	Submit preview frame (multipart)	S
GET	/face-eye/latest	Latest snapshot with age and stale flag	D, N
GET	/face-eye/frames/latest	Latest preview frame	D, N
GET	/face-eye/history	Snapshots by time range	D, N
GET	/face-eye/events	Monitoring events	D, N, A(R)
GET	/alerts	Alert list	A(R), D, N
POST	/alerts/:alertId/acknowledge	Acknowledge an alert	D, N
GET, POST	/patients/:patientId/tasks	List, create care tasks	A(R), D, N
PATCH, DELETE	/tasks/:taskId	Update, complete, delete a task	D, N
GET, POST	/patients/:patientId/medications	List, add medication entries	D, N
PATCH	/medications/:medId	Mark as given, edit	D, N
GET, POST	/patients/:patientId/notes	List, add clinical notes	D, N(R)
GET, PATCH	/beds	List beds; allocate, deallocate, set critical	A, D(R), N(R)
POST	/shifts/check-in, /shifts/check-out	Record shift start or end	D, N
GET, POST	/messages	Read, send messages	A, D, N, F
GET	/dashboard/summary	Counts, active sessions, recent events	A, D, N
GET	/system/health	Health check (details for admin)	P, A
GET	/audit-logs	Audit log	A

3.20. CV Payload Contract

Table 3.20 defines payload version 1. The first group is what the checked-in pipeline already sends (version `1.0`, no `schema_version` field). The second group requires the changes of CVI-02. Value ranges marked "to be confirmed" shall be fixed in the integration audit (Phase 5). An example is given in Appendix B.

Table 3.20.: CV feature payload, version 1

Field	Type	Unit / range	Req.	Current pipeline
Group 1 — fields in the checked-in payload				
patient_id	string	patient reference, e.g. P001	Yes	sent
timestamp_unix	number	seconds since epoch, UTC	Yes	sent
global_distress	number	0–100	Yes	sent
pain_index, fear_index, fatigue_index, agitation_index, tension_index, respiratory_index	number	0–100 each	Yes	sent
behavior_state	string	alert, drowsy, eye_closed, fixation, scanning	Yes	sent
gaze_yaw, gaze_pitch	number or null	radians	Yes	sent
gaze_variance	number	unit to be confirmed	Yes	sent
fixation_duration	number	seconds (30 FPS assumed in code)	Yes	sent
recent_saccades	integer	count, ≥ 0	Yes	sent
eye_contact_ratio	number	ratio	Yes	sent
episode_pain, episode_agitation, episode_fatigue, episode_fear, episode_eye_closure	boolean	—	Yes	sent
face_confidence	number	range to be confirmed	Yes	sent
Group 2 — additions (CVI-02)				
schema_version	string	e.g. "1.1"	Yes	new
units	object	declares gaze unit and similar	Should	new
baseline_ready	boolean	—	Should	in CSV/JSONL, not sent
fps, pipeline_version	number, string	measured frame rate; tag or commit	Should	new
seq	integer	increasing per session	Optional	new
au	object	AU4 to AU43 (15 values) and mar	Optional	computed, not sent
head_pose	object	yaw, pitch, roll in degrees	Optional	computed, not sent
motion_energy, facial_asymmetry	number	0–1; normalised	Optional	computed, not sent
ear_avg, eye_closed	number, boolean	eye aspect ratio; closure flag	Optional	computed, not sent
perclos	number	0–1	Optional	computed, not sent
mean_gaze_speed, max_gaze_speed	number	unit to be confirmed	Optional	in CSV, not sent

3.21. Data Requirements

MongoDB Atlas stores the data in the collections of Figure 3.3 and Table 3.21. The final schema may change with implementation; the requirements below remain.

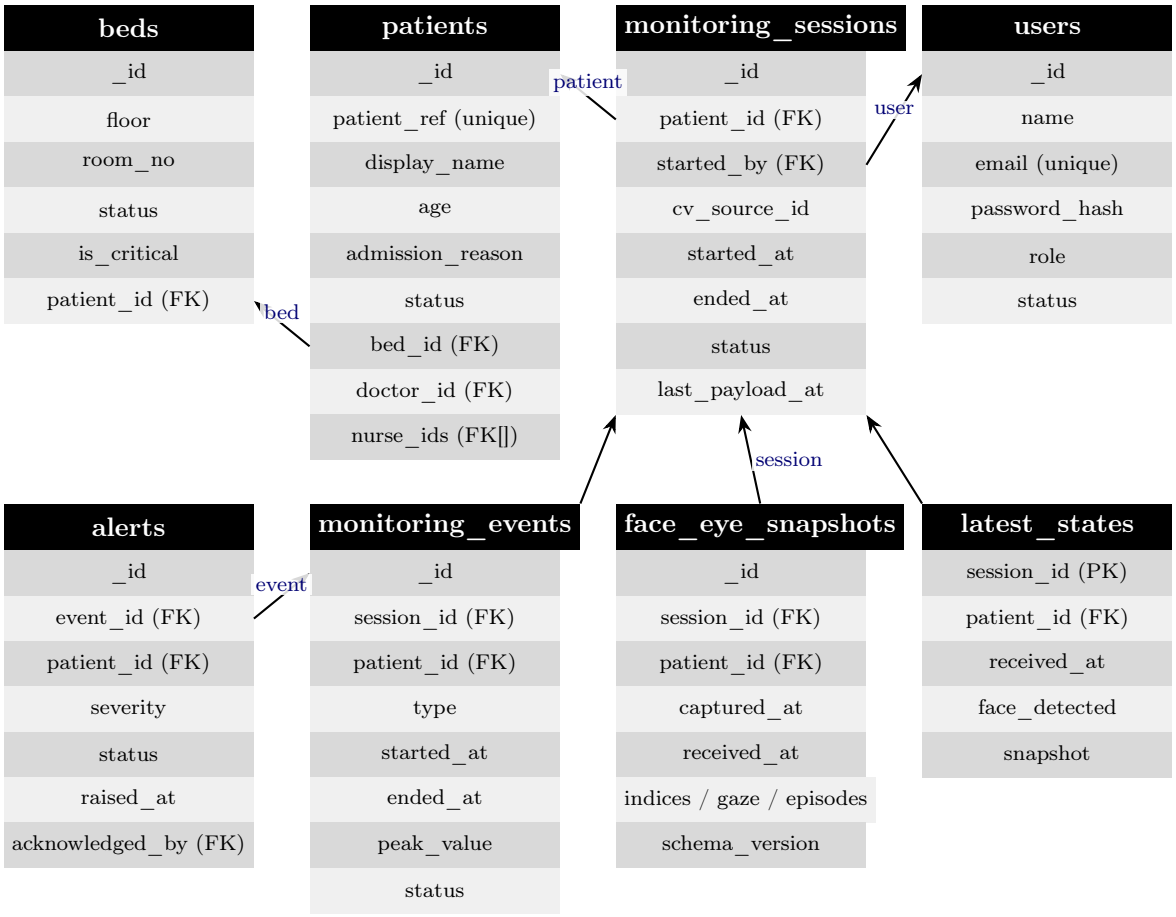


Figure 3.3.: Core data model (collections and foreign-key references). Administrative collections are listed in Table 3.21.

Table 3.21.: Collections, purpose and indexes

Collection	Purpose	Indexes and retention
users	Accounts, roles, status	unique email; role
patients	Minimal patient records	unique patient_ref; status; doctor_id; nurse_ids
beds	Floors, rooms, allocation	unique (floor, room_no); status
monitoring_sessions	One document per session	(patient_id, started_at desc); at most one active per patient
latest_states	Latest state per session	unique session_id
face_eye_snapshots	Time-ordered feature history	(session_id, received_at), (patient_id, received_at); TTL on received_at (default 30 days); time-series collection allowed

Collection	Purpose	Indexes and retention
monitoring_events	Events from episode flags	(session_id, started_at), (patient_id, started_at), status
alerts	Alerts and acknowledgements	(status, raised_at desc), (patient_id, raised_at)
care_tasks, medication_-schedules, clinical_notes	Administrative and clinical records	(patient_id, due_at); (patient_id, created_at)
shift_records, messages	Staff activity, messaging	(user_id, check_in_at); (thread, created_at)
audit_logs	Append-only audit trail	(user_id, at), (object, at); retention set by policy

Table 3.22.: Data requirements

ID	Statement
DR-01	Every document shall have a unique identifier and creation and update timestamps; times shall be stored in UTC.
DR-02	Frequently queried fields (patient references, session identifiers, timestamps, status) shall be indexed as in Table 3.21.
DR-03	The latest monitoring state shall be stored in `latest_states`, separate from historical snapshots, so that the live view never scans history.
DR-04	Each snapshot shall store the payload schema version and the fields received; fields not provided shall be stored as absent, not as zero.
DR-05	Snapshots shall expire through a TTL index (default 30 days, configurable). Events, alerts and clinical records follow a retention policy to be agreed with the supervisor and ethics adviser.
DR-06	Referential integrity shall be enforced in the application layer. Patients and users shall be archived or disabled rather than deleted, so that history keeps its references.
DR-07	Passwords shall be stored as hashes only. Video frames shall not be stored in the database.
DR-08	Only the fields listed in the schema shall be stored; the minimum-data rule of FR-PAT-06 applies to every collection.

4. Other Nonfunctional Requirements

Numeric values in this chapter are initial design targets (Section 1.2). They shall be measured during testing and reported as measured values.

4.1. Performance Requirements

These requirements keep the live view responsive and make sure that sending data to the platform does not slow the camera pipeline.

Table 4.1.: Performance requirements

ID	Requirement	Pri.
NFR-PERF-01	Under nominal load, a payload accepted by the backend shall appear on an open live monitor within 3 s at the 95th percentile (polling interval of 2 s included).	M
NFR-PERF-02	The feature-ingestion endpoint shall return its acknowledgement within 200 ms at the 95th percentile, excluding network time, so that the synchronous sender is not delayed.	M
NFR-PERF-03	Read endpoints for the latest state, alerts and dashboard summary shall respond within 500 ms at the 95th percentile; a history query for 24 hours of bucket-averaged data shall respond within 2 s.	S
NFR-PERF-04	The prototype shall support at least 5 concurrent active sessions and 25 concurrent dashboard users, each session posting at most 5 payloads per second.	S
NFR-PERF-05	The frontend shall load its first screen within 3 s on a broadband connection, use code splitting per workspace, and keep polling timers paused while the browser tab is hidden.	S
NFR-PERF-06	Enabling the backend connection shall not reduce the pipeline's processing rate measurably; the integration test shall compare frame rates with and without sending.	M

4.2. Security Requirements

Only registered users can enter the system, and each user can perform only the actions of their role.

Table 4.2.: Security requirements

ID	Requirement	Pri.
NFR-SEC-01	All traffic between browser, CV Service, backend and database shall use TLS; the backend shall send security headers (HSTS and the standard `helmet` set).	M
NFR-SEC-02	Passwords shall use bcrypt with cost at least 12, or Argon2id.	M

ID	Requirement	Pri.
NFR-SEC-03	Tokens shall be signed with a secret from the environment and be short-lived. Cookies shall be HttpOnly, Secure and SameSite, with protection against cross-site request forgery.	M
NFR-SEC-04	Authorisation shall be enforced on the server by role and by object (assignment), following the principle of least privilege.	M
NFR-SEC-05	All input shall be validated against a schema; query operators in user input shall be rejected to prevent NoSQL injection; output shall be encoded to prevent cross-site scripting; cross-origin access shall be restricted to the frontend origin.	M
NFR-SEC-06	Secrets (database URI, token secrets) shall be stored in environment variables and never committed to the repository; a `.env.example` shall document them.	M
NFR-SEC-07	The CV Service shall use its own service token, accepted only by the ingestion endpoints, revocable without affecting users.	M
NFR-SEC-08	Sign-in and ingestion endpoints shall be rate-limited.	S
NFR-SEC-09	The database user shall have only the permissions the backend needs, and the Atlas network access list shall be restricted to what the deployment requires.	S
NFR-SEC-10	Logs shall not contain passwords, tokens, video frames or free-text clinical content.	M
NFR-SEC-11	Dependencies shall be checked for known vulnerabilities (for example `npm audit`) before each release.	S

4.3. Privacy and Data Protection

The system observes patients by camera and stores clinical records, so privacy is a design requirement from the start.

Table 4.3.: Privacy requirements

ID	Requirement	Pri.
NFR-PRV-01	The system shall collect and store only the minimum patient information needed for the project (FR-PAT-06, DR-08).	M
NFR-PRV-02	Camera frames shall not be stored permanently by default (FR-ING-13).	M
NFR-PRV-03	Retention periods shall be configurable and documented (DR-05).	M
NFR-PRV-04	Access to patient data shall be limited by role and assignment and recorded in the audit log (FR-AUD-01).	S
NFR-PRV-05	Development and testing shall use volunteers, recorded or synthetic data. Real patient data shall be used only after ethics approval and consent (A-07).	M
NFR-PRV-06	Where possible the interface shall show the patient reference instead of the full name on shared screens.	C
NFR-PRV-07	The project shall not claim compliance with HIPAA, GDPR or similar regulations. The design supports future compliance work but does not establish it.	M

4.4. Reliability and Availability

A monitoring tool must make its own failures visible. These requirements cover failure handling and the honest display of missing data.

Table 4.4.: Reliability requirements

ID	Requirement	Pri.
NFR-REL-01	The platform shall degrade gracefully: if the CV Service is offline the dashboards shall show that state and the record-keeping features shall keep working.	M
NFR-REL-02	If the database is unavailable the API shall answer 503 with a clear error code and recover without restart when the database returns.	M
NFR-REL-03	Missing, stale and unavailable data shall be shown as such and never replaced by zero (FR-LIVE-12, DR-04).	M
NFR-REL-04	The backend shall handle missing faces, invalid frames, malformed payloads and disconnected services without crashing, and shall log them with a request identifier.	M
NFR-REL-05	The backend shall apply request timeouts (default 10 s) and the frontend shall show loading, empty and error states for every data view.	S
NFR-REL-06	The system shall offer a health endpoint for monitoring and automatic restart (FR-ING-17).	M
NFR-REL-07	Automated database backups shall be enabled where the Atlas plan supports them.	C

4.5. Usability and Accessibility

Doctors and nurses use the interface under time pressure, often on tablets.

Table 4.5.: Usability requirements

ID	Requirement	Pri.
NFR-USE-01	The interface shall be responsive from 360 px to 1920 px and usable by touch on tablets; touch targets shall be at least 44 px.	M
NFR-USE-02	The layout and wording shall be consistent across workspaces: one navigation pattern, one set of status chips, units shown next to every value.	S
NFR-USE-03	The interface shall target WCAG 2.1 level AA for colour contrast and keyboard operation of the main flows.	S
NFR-USE-04	Times shall be shown in the user's local time zone with the UTC value available on hover.	S
NFR-USE-05	A first-time doctor or nurse shall be able to open the live monitor of an assigned patient within three actions after sign-in.	S
NFR-USE-06	The interface language is English.	M

4.6. Maintainability and Portability

The code base is developed by a small team and extended after the project, so it must be easy to understand and reproduce.

Table 4.6.: Maintainability requirements

ID	Requirement	Pri.
NFR-MNT-01	The backend shall be organised in routes, controllers, services and models, with centralised error handling and request validation.	M
NFR-MNT-02	The repository shall use a linter and formatter, a README with set-up steps, and a `.env.example` for each component.	M
NFR-MNT-03	The API shall be documented in an OpenAPI description that matches Table 3.19 and Table 3.20.	S
NFR-MNT-04	Thresholds, intervals and retention periods named in this SRS shall be configuration values, not constants in code.	M
NFR-MNT-05	The frontend and backend shall run unchanged on any machine with the documented Node.js version and the environment variables set.	S

4.7. Software Quality Attributes

Quality is maintained by testing during development, not only at the end. Database, logical, API and user-interface tests are required, and each requirement in this document is written to be verifiable.

Table 4.7.: Testing requirements

ID	Requirement	Pri.
NFR-TST-01	Unit tests shall cover validation, authorisation rules, event derivation, freshness and stale logic, and the data-access layer.	M
NFR-TST-02	API tests shall cover every endpoint of Table 3.19 for success, validation errors, 401, 403, 404 and 409 cases.	M
NFR-TST-03	A contract test shall check that the payload of Table 3.20 accepted by the backend is the same one the frontend renders, so that a field-name mismatch cannot go unnoticed.	M
NFR-TST-04	A CV simulator shall replay recorded or synthetic payloads (including no-face gaps, an eye-closed period and a stopped service) against the API so that the full flow can be tested without a camera.	M
NFR-TST-05	Integration tests shall cover the whole path camera or simulator → API → MongoDB → dashboard, including camera disconnect, no face, CV Service stopped and database unavailable.	M
NFR-TST-06	UI tests shall cover sign-in per role, workspace guards, live monitor states (live, stale, offline, no data) and alert acknowledgement.	S
NFR-TST-07	A load test shall check the targets of NFR-PERF-01 to NFR-PERF-04 and the results shall be recorded.	S
NFR-TST-08	Any statement about CV accuracy requires annotated evaluation data and a separate experimental method. Successful integration alone shall not be presented as clinical validity.	M

4.8. Business Rules

The platform stores monitoring information and supports staff; it does not make clinical decisions. These rules govern every screen, report and demonstration.

Table 4.8.: Business rules

ID	Statement
BR-01	The system is a research and monitoring-support prototype. It is not a validated medical device and does not diagnose, prescribe or replace physiological measurement or professional judgement.
BR-02	Every index, behaviour state, severity label and episode shall be presented as a visual heuristic derived from facial and eye features, with the software-defined thresholds of the pipeline.
BR-03	No value shall be fabricated. A module without a data source is hidden or marked "Not available in this version". Illustrative content is labelled as sample data.
BR-04	Tasks and medication entries are administrative records. The system shall not recommend, schedule automatically or alter treatment.
BR-05	Only authorised staff responsible for a patient may acknowledge that patient's alerts.
BR-06	Camera frames are live previews and are not stored unless a documented retention feature is enabled.
BR-07	Changes to severity thresholds, alert rules and retention periods are configuration changes and shall be recorded in the project documentation.
BR-08	Patient and user records are archived or disabled, never deleted through the interface.
BR-09	Accuracy or performance claims about the CV indices shall be made only with annotated evaluation data (NFR-TST-08).

5. Other Requirements

The ICU Monitoring Platform will need maintenance, because the requirements will change as the CV pipeline is audited and the project scope is confirmed. This chapter covers deployment, legal and ethical conditions, deliverables, future scope and the known issues that must be resolved before the system is considered integrated.

5.1. Deployment

Figure 5.1 shows the target deployment. The frontend and backend are deployed from the same GitHub repository to Vercel and Render; the database is a managed MongoDB Atlas cluster; the CV Service stays on the machine that has the camera and sends data outward.

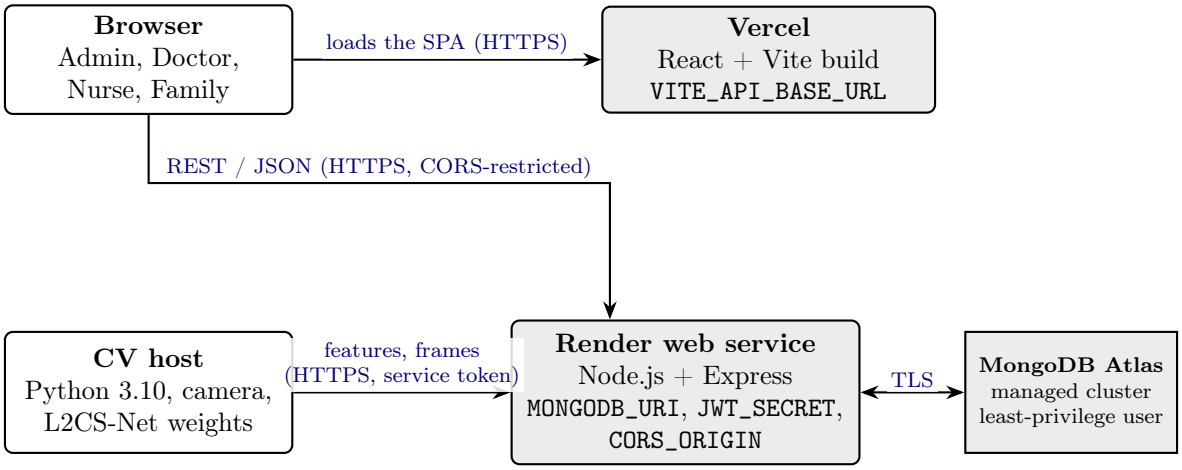


Figure 5.1.: Deployment view.

Table 5.1.: Main configuration values

Variable	Component	Purpose
VITE_API_BASE_URL	Frontend	Base URL of the backend API.
MONGODB_URI	Backend	Atlas connection string (secret).
JWT_SECRET	Backend	Signing secret for user tokens (secret).
CORS_ORIGIN	Backend	Allowed frontend origin.
CV_SERVICE_TOKEN	Backend	Accepted service token for ingestion (secret; stored hashed).
STALE_AFTER_S, OFFLINE_AFTER_S	Backend	Freshness thresholds (default 5 s and 30 s).
SNAPSHOT_INTERVAL_S, SNAPSHOT_TTL_DAYS	Backend	History rate and expiry (default 1 s and 30 days).
ALERT_COOLDOWN_S	Backend	Alert suppression window (default 60 s).
BACKEND_URL, SERVICE_TOKEN, ENABLE_BACKEND	CV Service	Target API, token and on/off switch of the sender.
SEND_RATE_HZ, CAMERA_INDEX	CV Service	Post rate (default 2) and camera device.

5.2. Legal, Ethical and Regulatory Conditions

- Any use of the system with real patients or volunteers needs approval from the responsible ethics body, informed consent, and a data-handling plan (A-07, NFR-PRV-05).
- The project does not claim regulatory approval or compliance (FR-LAND-04, NFR-PRV-07).
- Third-party components keep their licences: MediaPipe, OpenCV, PyTorch and the L2CS-Net weights and code used by the pipeline. The licence of the weights file shall be checked before redistribution.

5.3. Documentation Deliverables

Technical documentation, architecture diagrams, the OpenAPI description, a test report, screenshots and a demonstration of the complete monitoring workflow shall be delivered together with the source code and the final project report. The report shall say what is implemented, what is a heuristic, what is planned and what is not validated.

5.4. Maintenance and Future Scope

Likely extensions (not part of Version 1.0): secure real-time video streaming; additional sensors for multimodal fusion; improved temporal modelling and use of the trained model that already exists in the repository; clinician-annotated datasets for validation; alert rules that can be edited in the interface; better buffering and monitoring of the sender; role-specific accessibility work.

5.5. Known Issues and Risks

The items below come from the static review of the repository and from the project proposal. They are risks to integration, not defects of this specification.

Table 5.2.: Known issues and risks

ID	Issue or risk	Handled by
R-01	The sender posts to `/api/face-eye/features` and `/api/stream/face`; the in-repo FastAPI service defines only `/face`, `/posture` and `/fusion`, so the two do not match.	CVI-01, FR-ING-01
R-02	The sender is disabled by default, does not check HTTP status and has no retry; failures are silent.	CVI-04, CVI-06
R-03	The 1 s synchronous send inside the camera loop can stall inference.	CVI-03, NFR-PERF-02, NFR-PERF-06
R-04	The payload omits AUs, head pose, PERCLOS and gaze speed that the proposal's dashboard shows.	CVI-02, FR-LIVE-10
R-05	No heartbeat: "no face in view" cannot be told from "CV Service stopped".	CVI-06, FR-LIVE-13

ID	Issue or risk	Handled by
R-06	The in-repo backend keeps only the latest values in RAM, has no history, no authentication and no frame route.	Replaced by this platform
R-07	An earlier frontend showed incorrect data and its cause was not recorded; its source is not in the repository.	NFR-TST-03 (contract test)
R-08	Comments and default values of the AU calibration in `config/settings.py` appear inconsistent (for example the AU6 range).	Audit with neutral-face logs before AU values are shown as results
R-09	Gaze durations assume 30 FPS; the saccade window comment (150 frames) differs from its description; gaze angles are in radians.	CVI-09, FR-ING-14
R-10	In the eye-closed branch the gaze tracker returns zeroed variance and speed fields.	FR-LIVE-07
R-11	The JSONL log lacks gaze and behaviour state; the CSV is overwritten at each run while the JSONL is appended.	CVI-11; note for experiments
R-12	Root `requirements.txt` and `requirements/` differ (CUDA pins, duplicate OpenCV packages); the README mentions an `L2CS-Net` folder that does not exist.	One tested install path per target (Windows, Jetson, Docker)
R-13	The indices are not clinically validated; a trained XGBoost model exists but `main.py` does not use it.	BR-01, BR-02, BR-09
R-14	Free-tier hosting limits (idle spin-down, storage, no GPU).	C-04, C-12, DR-05
R-15	Camera and patient data are sensitive.	NFR-PRV-01 to NFR-PRV-07
R-16	The pipeline was reviewed statically and has not been run end-to-end against the platform.	A-01; integration tests NFR-TST-05

A. Traceability Matrix

Table A.1.: Objectives, proposal objectives, requirements and verification

Obj.	Objective (Section 1.5)	Requirements	Proposal obj. / verification
O1	Integrate the existing pipeline through a versioned contract	FR-ING, CVI, Table 3.20	1, 5 / NFR-TST-03, -05
O2	Secure role-based access	FR-AUTH, FR-WS, NFR-SEC	7 / NFR-TST-02, -06
O3	Persist data in MongoDB Atlas	DR-01 to DR-08, FR-PAT, FR-SESS	6 / NFR-TST-01, -05
O4	Live view with explicit stale and missing states	FR-LIVE, FR-ING-07, NFR-PERF-01, NFR-REL-03	8 / NFR-TST-05, -07
O5	Events and alerts with acknowledgement	FR-EVT	9 / NFR-TST-01, -06
O6	Historical charts and summaries	FR-HIST, FR-ING-08	9 / NFR-TST-02
O7	Rooms, shifts, tasks, medication records	FR-ADM, FR-NUR, FR-DOC	9 / NFR-TST-02, -06
O8	Responsive interface modelled on the reference video	Section 2.8, IR-UI, NFR-USE	2, 3 / NFR-TST-06
O9	Deployment on Vercel, Render and Atlas	Section 5.1, C-01 to C-04	4, 10 / deployment test
O10	Testing and documentation without clinical claims	Chapter 4, BR, Section 5.3	11, 12 / NFR-TST-08

B. Example Payload

The listing shows the shape of a version 1.1 payload. All values are illustrative only and are not measurements.


```
{
  "schema_version": "1.1",
  "units": { "gaze": "rad", "head_pose": "deg" },
  "patient_id": "P001",
  "timestamp_unix": 1760000000.123,
  "behavior_state": "alert",
  "global_distress": 12.0,
  "pain_index": 8.0, "fear_index": 5.0, "fatigue_index": 20.0,
  "agitation_index": 3.0, "tension_index": 10.0, "respiratory_index": 4.0,
  "gaze_yaw": 0.05, "gaze_pitch": -0.02, "gaze_variance": 0.01,
  "fixation_duration": 0.8, "recent_saccades": 2, "eye_contact_ratio": 0.7,
  "episode_pain": false, "episode_agitation": false, "episode_fatigue": false,
  "episode_fear": false, "episode_eye_closure": false,
  "face_confidence": 0.9,
  "baseline_ready": true, "fps": 24.5, "pipeline_version": "5405f41",
  "head_pose": { "yaw": 3.0, "pitch": -2.0, "roll": 1.0 },
  "perclos": 0.1
}
```

C. Open Decisions

Table C.1.: Decisions to confirm with the supervisor and team

ID	**Statement**
D-01	Final role set and whether the Family workspace belongs in Version 1.0.
D-02	Where the CV Service runs (laboratory PC, Jetson device or cloud GPU) and how its token is provisioned.
D-03	Retention periods for snapshots, events and clinical records, and the ethics approval needed.
D-04	Polling or Server-Sent Events/WebSockets for the live view, based on the measured latency.
D-05	Alert thresholds and cool-down values, and who may change them.
D-06	Token delivery (HttpOnly cookie or bearer header) and the idle time-out.
D-07	Whether live preview video is required for the demonstration, and the bandwidth limit.
D-08	Size of the seeded room and floor data (the reference video shows five floors of 40 rooms).
D-09	JavaScript or TypeScript for frontend and backend.
D-10	Annotated dataset and metrics for any evaluation of CV accuracy; this is separate from the website integration.